

1. Variables & Scoping What is the difference between var, let, and const in JavaScript?

Ans: var is global or function scope but let, const scope is in block.

2. Operators What is the difference between == and ===? Give one example.

Ans: The equal “==” loosely compares equality. Only compares value. On the other hand, “===” strictly compares value and type also.

Example

5 == “5” **true** (compare value only)

5 === “5” **False** (Compare value and type)

3. Loops What is the difference between a for loop and a for-of loop?

Ans: general for loop easy to get index need to travel array using index number like array[i], for...of loop easier to access traverse array items it's directly return items.

4. Functions What is an arrow function? Write a simple example.

Ans: arrow function is a modern JavaScript syntaxe that use arrow sign => to write function. Example:

```
const add = (a, b) => {  
  return a + b;  
};
```

5. Advanced Functions What are an IIFE, a recursive function, and a generator function? Write one key characteristic of each.

Ans:

IIFE - immediately invoked function expression. This function called immediately on initialisation no need to call this type of function to operation.

Key characteristics: call on initialization

Recursive function - this type of function call itself again.

Example:

```
function countdown(n) {  
  if (n === 0) return;  
  
  console.log(n);  
  countdown(n - 1);  
}  
countdown(3);
```

Key characteristics: call it self

Generator: this is a special function using function* and yield keyword we can define this function.

Example:

```
function* numbers() {  
  yield 1;  
  yield 2;  
  yield 3;  
}
```

```
const gen = numbers();
```

```
console.log(gen.next().value); // return 1
```

```
console.log(gen.next().value); // return 2
```

```
console.log(gen.next().value); // return 3
```

Key characteristics: can pause and resume